

## Project Initialization and Planning Phase

|               |                |
|---------------|----------------|
| Date          | 30 June 2026   |
| Team ID       | XXXXXXXX       |
| Project Title | Raising waters |
| Maximum Marks | 5 Marks        |

### Project Proposal (Proposed Solution) report

The proposed solution aims to improve flood prediction using machine learning techniques to provide accurate and timely flood risk assessment. By analyzing historical weather data such as rainfall, cloud visibility, and seasonal rainfall patterns, the system predicts the likelihood of flood events. The solution enhances disaster preparedness, supports early warning systems, and helps authorities make informed decisions to reduce the loss of life and property.

| Project Overview  |                                                                                                                                                                                                                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to develop a machine learning-based flood prediction system that accurately predicts flood occurrences and provides timely alerts to support disaster management and public safety.                                                                                                                         |
| Scope             | The project focuses on analyzing historical weather data, comparing multiple machine learning algorithms, selecting the best-performing model, and integrating it into a Flask web application for real-time flood risk prediction. The application is designed for deployment on IBM Cloud to ensure scalability and accessibility. |
| Problem Statement |                                                                                                                                                                                                                                                                                                                                      |
| Description       | Conventional flood forecasting methods often provide delayed or inaccurate predictions, making it difficult for authorities to take timely preventive actions. The absence of reliable early warning systems increases the impact of floods on communities and infrastructure.                                                       |

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|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Impact                   | An accurate flood prediction system enables early warnings, supports better disaster response planning, minimizes property damage, and helps protect human lives by allowing authorities to act before flood situations become critical.                                                                                                                                                                                                                                                                                                             |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Approach                 | The solution uses machine learning algorithms including <b>Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost</b> to analyze historical weather data and predict flood risk. The best-performing model is integrated into a Flask web application, allowing users to enter weather parameters and receive instant flood predictions.                                                                                                                                                                                              |
| Key Features             | <ul style="list-style-type: none"><li><input type="checkbox"/> Machine learning-based flood prediction.</li><li><input type="checkbox"/> Comparison of multiple classification algorithms.</li><li><input type="checkbox"/> XGBoost model with <b>96.55% prediction accuracy</b>.</li><li><input type="checkbox"/> Simple Flask web application for flood risk prediction.</li><li><input type="checkbox"/> Early warning support for disaster management.</li><li><input type="checkbox"/> IBM Cloud deployment for remote accessibility.</li></ul> |

Resource Requirements

| Resource Type       | Description                             | Specification/Allocation   |
|---------------------|-----------------------------------------|----------------------------|
| <b>Hardware</b>     |                                         |                            |
| Computing Resources | CPU/GPU specifications, number of cores | Intel Core i5/i7 Processor |
| Memory              | RAM specifications                      | 8 GB                       |
| Storage             | Disk space for data, models, and logs   | 256 GB SSD (or higher)     |
| <b>Software</b>     |                                         |                            |
| Frameworks          | Python frameworks                       | Flask                      |

|                         |                      |                                                  |
|-------------------------|----------------------|--------------------------------------------------|
| Libraries               | Additional libraries | scikit-learn, pandas, numpy, matplotlib, seaborn |
| Development Environment | IDE                  | Jupyter Notebook, VS Code, PyCharm               |
| <b>Data</b>             |                      |                                                  |
| Data                    | Source, size, format | Kaggle dataset, 614, csv UCI dataset, 690, csv   |